Spectrum of a Ring and Structure Sheaf

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We cover the following topics in this note.

- Spectrum of a Ring
- Sheaf

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1 Observations

$$\begin{tikzcd} \mathbb{C}[x] \arrow[d, shift right, "\phi_\alpha"'] \\ \mathbb{C} \arrow[u, shift right, "f^{-1}"'] \end{tikzcd} \qquad \begin{tikzcd} \{\langle 0\rangle\} \cup \{\langle x-\alpha\rangle \mid \alpha \in \mathbb{C}\} \\ \{\langle 0\rangle\} \arrow[u] \end{tikzcd}$$

- Rings: $\mathbb{C}[x], \mathbb{C}$;
- The **evaluation homomorphism** at $x = \alpha$:

$$\begin{array}{cccc} \phi_\alpha & : & \mathbb{C}[x] & \longrightarrow & \mathbb{C} \\ & & p(x) & \longmapsto & p(\alpha) \end{array}$$

Spectrum of Ring

CRing: Category of Commutative Rings

- **Objects**: commutative rings $R, S, \ldots$
- **Morphisms**: ring homomorphisms $f : R \to S, \ldots$

Top: The Category of Topological spaces

- **Objects**: topological spaces $(X, \tau_X), (Y, \tau_Y), \ldots$
- **Morphisms**: continuous maps $f : (X, \tau_X) \to (Y, \tau_Y)$, which satisfy: $U \in \tau_Y \Rightarrow f^{-1}[U] \in \tau_X$.

The Functor $\mathrm{Spec} : \mathbf{CRing}^{\mathrm{op}} \to \mathbf{Top}$

- **Objects**
  - For a commutative ring R, $\mathrm{Spec}(R)$ is the *spectrum of* R:

$$\mathrm{Spec}(R) = \left\{ \mathfrak{p} \subseteq R : \mathfrak{p} \text{ is a prime ideal} \right\}.$$

  - An ideal $\mathfrak{p} \trianglerighteq R$ is prime if

$$ab \in \mathfrak{p} \implies a \in \mathfrak{p} \text{ or } b \in \mathfrak{p}.$$

  - *Zariski Topology* on $\mathrm{Spec}(R)$:
    The spectrum $\mathrm{Spec}(R)$ is quipped with **Zariski topology**, where the closed sets are

$$V(I) = \left\{ \mathfrak{p} \in \mathrm{Spec}(R) : I \subseteq \mathfrak{p} \right\}$$

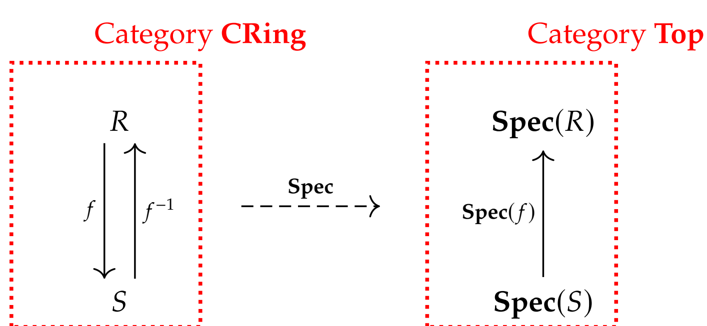
    for any ideal $I \trianglerighteq R$.

- **Morphisms**

  For a ring homomorphism $f : R \to S$ in **CRing**, the corresponding morphism in **Spec** is a continuous map

$$\begin{array}{rcrcl} \mathbf{Spec}(f) & : & \mathbf{Spec}(S) & \longrightarrow & \mathbf{Spec}(R) \\ & & \mathfrak{p} & \longmapsto & f^{-1}[\mathfrak{p}] \end{array},$$

  where $f^{-1}[\mathfrak{p}] = \left\{ r \in R : f(r) \in \mathfrak{p} \right\}$ is the preimage of the prime ideal $\mathfrak{p} \in \mathbf{Spec}(S)$.



2 Category, Functor and Natural Transformation

$$
\begin{tikzcd}
A \arrow[r, "f"] \arrow[dr, "g\circ f"'] \arrow[loop left, "1_A"] & B \arrow[d, "g"] \arrow[loop right, "1_B"] \\
 & C \arrow[loop below, "1_C"]
\end{tikzcd}
$$

Figure 1: Category

Category

Definition 1. A **category** $\mathcal{C}$ consists of the following data:

(i) (**Objects**) A class $\boxed{\mathrm{Ob}(\mathcal{C})}$, whose elements are called **objects** of category $\mathcal{C}$.

(ii) (**Morphisms**) For each pair of objects $A, B \in \mathrm{Ob}(\mathcal{C})$, a class $\boxed{\mathrm{Mor}_{\mathcal{C}}(A, B)}$[a], whose elements are called **morphisms** (or **arrows**) from A to B.

(iii) (**Composition**) For any $f \in \mathrm{Mor}_{\mathcal{C}}(A, B)$ and $g \in \mathrm{Mor}_{\mathcal{C}}(B, C)$, a **composition**

$$\boxed{(g \circ f) \in \mathrm{Mor}_{\mathcal{C}}(A, C)}.$$

These data are to satisfy the following axioms:

1. (**Identity**) For every object $A \in \mathrm{Ob}(\mathcal{C})$, there exists an **identity morphisms**

$$\boxed{1_A \in \mathrm{Mor}_{\mathcal{C}}(A, A)}.$$

2. (**Associativity**) For any $f \in \mathrm{Mor}_{\mathcal{C}}(A, B)$, $g \in \mathrm{Mor}_{\mathcal{C}}(B, C)$, and $h \in \mathrm{Mor}_{\mathcal{C}}(C, D)$,

$$\boxed{h \circ (g \circ f) = (h \circ g) \circ f}.$$

[a] Sometimes, we also denote it by $\mathrm{Hom}(A, B)$ or $\mathcal{C}(A, B)$.

Remark 1.

$$\begin{tikzcd} A \arrow[r, dashed, "f"] \arrow[dr, "g\circ f"'] & B \arrow[d, dashed, "g"] \\ & C \end{tikzcd}$$

Figure 2: Composition of Morphisms

$$\begin{tikzcd} A \arrow[loop above, "1_A"] \arrow[r, shift left, "f=f\circ 1_A"] & B \arrow[l, shift left, "g=1_A\circ g"] \arrow[loop below, "1_B"] \end{tikzcd}$$

Figure 3: Identity Morphisms

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[rr, bend left, "g\circ f"] \arrow[rrr, bend left=60, "h\circ(g\circ f)"] \arrow[rrr, bend right=60, "(h\circ g)\circ f"'] & B \arrow[r, "g"] \arrow[rr, bend right, "h\circ g"'] & C \arrow[r, "h"] & D \end{tikzcd}$$

Figure 4: Associativity of Composition

Example 1.

Category	Object	Morphism
$\mathscr{S}$et	Sets	Functions between sets
$\mathscr{G}$roup	Groups	Group Homomorphisms
$\mathscr{R}$ing	Rings	Ring Homomorphisms
$\mathscr{V}$ector Space	Vector Spaces	Linear Transformations
$\mathscr{T}$opology Space	Topological Spaces	Continuous Maps
$\mathscr{P}$oset	Partially Ordered Sets	Order-preserving Maps
$\vdots$	$\vdots$	$\vdots$

Functor

Definition 2. A **functor** $\mathbf{F} : \mathcal{C} \to \mathcal{D}$ is a mapping between two categories $\mathcal{C}$ and $\mathcal{D}$ is given by the following data:

(i) (**Object Mapping**) A mapping that assign to each object $A \in \mathrm{Ob}(\mathcal{C})$ an object $\mathbf{F}(A) \in \mathrm{Ob}(\mathcal{D})$:

$$\begin{array}{cccc} \mathbf{F} & : & \mathrm{Ob}(\mathcal{C}) & \longrightarrow & \mathrm{Ob}(\mathcal{D}) \\ & & A & \longmapsto & \mathbf{F}(A) \end{array}$$

(ii) (**Morphism Mapping**) For each pair $A, B \in \mathrm{Ob}(\mathcal{C})$, a mapping that assigns to each morphism $\phi \in \mathrm{Mor}_{\mathcal{C}}(A, B)$ a morphism $\mathbf{F}(\phi) \in \mathrm{Mor}_{\mathcal{D}}(\mathbf{F}(A), \mathbf{F}(B))$:

$$\begin{array}{ccccc} \mathbf{F} & : & \mathrm{Mor}_{\mathcal{C}}(A, B) & \longrightarrow & \mathrm{Mor}_{\mathcal{D}}(\mathbf{F}(A), \mathbf{F}(B)) \\ & & \phi & \longmapsto & \mathbf{F}(\phi) \end{array}$$

Remark 2 (Compatibility with Identity Morphisms and Composition)**.** These functors satisfy the following conditions:

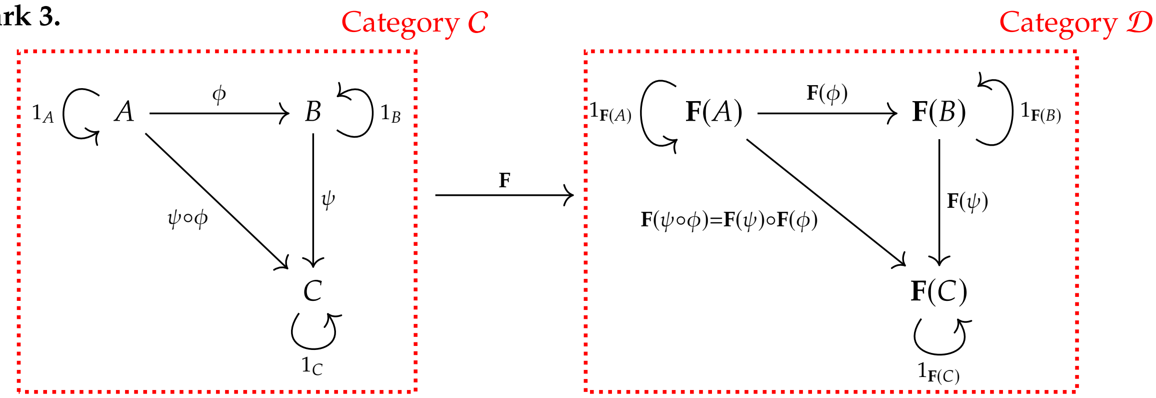
1. (**Preservation of Identity Morphisms**) For every object $A \in \mathrm{Ob}(\mathcal{C})$,

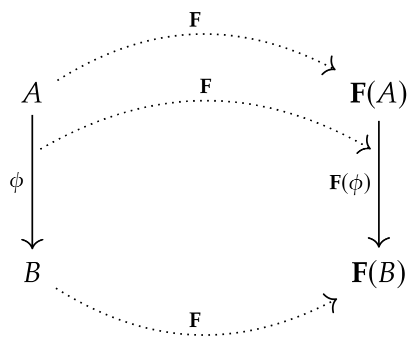
$$\mathbf{F}(1_A) = 1_{\mathbf{F}(A)}.$$

2. (**Preservation of Composition**) For all morphisms $\phi \in \mathrm{Mor}_{\mathcal{C}}(A, B)$ and $\psi \in \mathrm{Mor}_{\mathcal{C}}(B, C)$,

$$\mathbf{F}(\psi \circ \phi) = \mathbf{F}(\psi) \circ \mathbf{F}(\phi).$$

Remark 3.





Example 2.

Category (Source)	Functor	Object Mapping	Morphism Mapping	Category (Target)
TBA	TBA	TBA	TBA	TBA
⋮	⋮	⋮	⋮	⋮

Natural Transformation

Definition 3. Let $\mathcal{C}$ and $\mathcal{D}$ be categories, and let $\mathbf{F}, \mathbf{G} : \mathcal{C} \to \mathcal{D}$ be two functors. A **natural transformation** (or a **morphism of functors**) $\eta : \mathbf{F} \Rightarrow \mathbf{G}$, is a collection $\{\eta_A\}_{A \in \mathrm{Ob}(\mathcal{C})}$ such that

(i) (**Component Morphisms**) For each object $A \in \mathrm{Ob}(\mathcal{C})$, there is a morphism

$$\eta_A : \mathbf{F}(A) \longrightarrow \mathbf{G}(A)$$

in the category $\mathcal{D}$, called the **component of** η **at** A.

(ii) (**Naturality Condition**) For each morphism $\phi : A \to B$ in $\mathcal{C}$ the following diagram in $\mathcal{D}$ commutes:

$$\begin{tikzcd}
\mathbf{F}(A) \arrow[r, "\eta_A"] \arrow[d, "\mathbf{F}(\phi)"'] & \mathbf{G}(A) \arrow[d, "\mathbf{G}(\phi)"] \\
\mathbf{F}(B) \arrow[r, "\eta_B"] & \mathbf{G}(B)
\end{tikzcd}$$

In other words,

$$\eta_B \circ \mathbf{F}(\phi) = \mathbf{G}(\phi) \circ \eta_A$$

for all $\phi : A \to B$ in $\mathcal{C}$.

Remark 4. Sometimes we use the diagram

$$\begin{tikzcd}
\mathcal{C} \arrow[r, bend left, "\mathbf{F}"] \arrow[r, bend right, "\mathbf{G}"'] \arrow[r, phantom, "\Downarrow\eta"] & \mathcal{D}
\end{tikzcd}$$

to indicate that η is a morphism form $\mathbf{F}$ to $\mathbf{G}$.

Remark 5.

$$
\begin{tikzcd}
A \arrow[r, dotted, bend left, "\mathbf{F}"] \arrow[rr, dotted, bend left=50, "\mathbf{G}"] \arrow[d, "\phi"'] \arrow[r, phantom, "\cdots\mathbf{F}\cdots" description] & \mathbf{F}(A) \arrow[r, "\eta_A"] \arrow[d, "\psi=\mathbf{F}(\phi)"] & \mathbf{G}(A) \arrow[d, "\chi=\mathbf{G}(\phi)"] \\
B \arrow[r, dotted, bend right, "\mathbf{F}"'] \arrow[rr, dotted, bend right=50, "\mathbf{G}"'] & \mathbf{F}(B) \arrow[r, "\eta_B"] & \mathbf{G}(B)
\end{tikzcd}
$$

Example 3 (Determinant)**.** We start with the basic concepts and gradually introduce the determinant as a mathematical object.

Step 1. Understanding Matrices.

A matrix is a rectangular array of numbers. For example, a 2×2 matrix look like this:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

where a, b, c, d are numbers.

Step 2. Invertibility of Matrices. The inverse of $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{R})$ (if it exists) is given by:

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

The determinant is a single number associated with a matrix, defined for 2×2 matrices as:

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

Step 3. The Determinant as a Function from $M_2(\mathbb{R})$ to $\mathbb{R}$

The set $M_2(\mathbb{R})$ is the set of all 2×2 matrices with real entries:

$$M_2(\mathbb{R}) := \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{R} \right\}.$$

The determinant is a function:

$$\begin{array}{rccl} \det & : & M_2(\mathbb{R}) & \longrightarrow & \mathbb{R} \\ & & A & \longmapsto & \det(A) = ad - bc \in \mathbb{R} \end{array}$$

for each $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{R})$. This function assigns a real number to each 2×2 matrix.

Step 4. Properties of the Determinant Function.

- (**Multiplicativity**) For any $A, B \in M_2(\mathbb{R})$,

  $$\det(A * B) = \det(A) \cdot \det(B)$$

  with $*$ is matrix multiplication and $\cdot$ is number multiplication.

- (**Normalization**) The determinant of the identity matrix is:

  $$\det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1.$$

Step 5. Group Structure

We identify a natural subset of $M_2(\mathbb{R})$ where:

- The determinant is nonzero.
- The operation of matrix multiplication is preserved.

1. (Closure Under Multiplication) For A, B in this subset, their product is also a 2×2 matrix. Using the determinant property,

   $$\det(A) \neq 0 \wedge \det(B) \neq 0 \implies \det(AB) = \det(A)\det(B) \neq 0.$$

2. (Associativity) Matrix multiplication is associative.
3. (Existence of Identity and Inverses) The identity matrix $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ satisfies $\det(I) = 1 \neq 0$, and for every matrix A in this subset, there exists A^{-1} since $\det(A) \neq 0$.

These properties define a group, and this subset of matrices is precisely the general linear group of invertible matrices:

$$\mathrm{GL}(2,\mathbb{R}) := \left\{A \in M_2(\mathbb{R}), \det(A) = ad - bc \neq 0\right\}.$$

Step 6. The Determinant as a Group Homomorphism from GL$(2,\mathbb{R})$ to $\mathbb{R}^\times$

The determinant is restricted to invertible matrices:

$$\det : \mathrm{GL}(2,\mathbb{R}) \longrightarrow \mathbb{R}^\times,$$

where $\mathbb{R}^\times$ is the set of nonzero real numbers under multiplication. The determinant preserves the group operation because because of the multiplicative property:

$$\det(A * B) = \det(A) \cdot \det(B).$$

for all $A, B \in \mathrm{GL}(2,\mathbb{R})$.

Step 7. Matrices Over a Commutative Ring

Let R be a commutative ring. The determinant of $A \in M_2(R)$ is define as

$$\det A = a \cdot d + (-(b \cdot c)),$$

using the addition and multiplication operations of the commutative ring R.

Step 8. Categories

- The Category CRing: The Category of Commutative Rings
  - Objects: commutative rings
  - Morphisms: ring homomorphisms
  - Composition: standard composition of functions
  - Identity Morphisms: $1_R : R \longrightarrow R$ defined by $1_R(a) = a$ for $a \in R$
- The Category Grp: The Category of Groups
  - Objects: groups
  - Morphisms: group homomorphisms
  - Composition: standard composition of functions
  - Identity Morphisms: $1_G : G \longrightarrow G$ defined by $1_G(a) = a$ for $a \in G$

Step 9. Functors

- The General Linear Group Functor $\mathbf{GL}_2$
  - $R \mapsto \mathbf{GL}_2(R)$
  - $(\phi : R \to S) \mapsto (\mathbf{GL}_2(\phi) : \mathbf{GL}_2(R) \to \mathbf{GL}_2(S))$
- The Unit Functor $\mathbf{U}$
  - $R \mapsto \mathbf{U}(R) = R^\times$
  - $(\phi : R \to S) \mapsto (\mathbf{U}(\phi) : R^\times \to S^\times)$

Step 10. Determinant as Natural Transformation

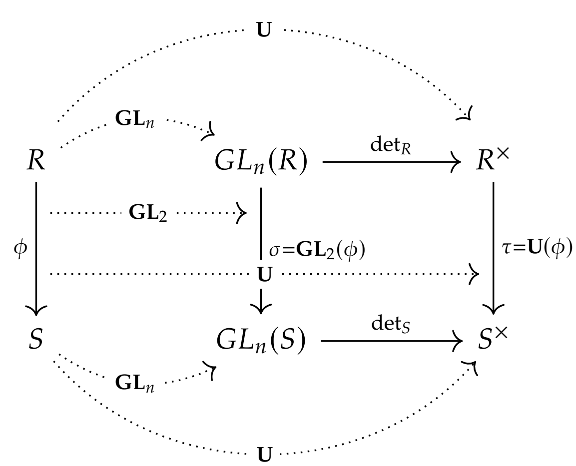
The determinant map is a family of group homomorphisms

$$\{\eta_R\}_{R \in \mathrm{Ob(CRing)}}, \quad \text{where} \quad \eta_R : \mathbf{GL}(2, R) \to R^\times,$$

one for each ring R, defined by

$$\eta_R(A) = \det A.$$

$$\begin{tikzcd}
R \arrow[d, "\phi"'] & GL_n(R) \arrow[r, "\eta_R=\det_R"] \arrow[d, "\mathbf{GL}_n(\phi)"] & R^\times \arrow[d, "\mathbf{U}(\phi)"] \\
S & GL_n(S) \arrow[r, "\eta_S=\det_S"] & S^\times
\end{tikzcd}$$



3 Presheaf and Sheaf

Presheaf

Definition 4. Let X be a topological space. A **presheaf** $\mathcal{F}$ of abelian groups on X consists of

(i) For each open $U \subseteq X$, an abelian group $\mathcal{F}(U)$, elements of $\mathcal{F}(U)$ are called sections of $\mathcal{F}$ over U.

(ii) For each $V \subseteq U$ a homomorphism (called restriction map)

$$\rho_{U,V} : \mathcal{F}(U) \to \mathcal{F}(V)$$

such that $\rho_{U,U}$ is an identity and for open sets $W \subseteq V \subseteq U$ we have

$$\rho_{U,W} = \rho_{V,W} \circ \rho_{U,V}.$$

Remark 6.

$$\begin{tikzcd}
V \arrow[rr] & & U \\
& W \arrow[ul] \arrow[ur] &
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathcal{F}(V) \arrow[dr, "\rho_{V,W}"'] & & \mathcal{F}(U) \arrow[ll, "\rho_{U,V}"'] \arrow[dl, "\rho_{U,W}"] \\
& \mathcal{F}(W) &
\end{tikzcd}$$

Sheaf

Definition 5. TBA